PROJECT 1

BHARAT MOBILITY POD

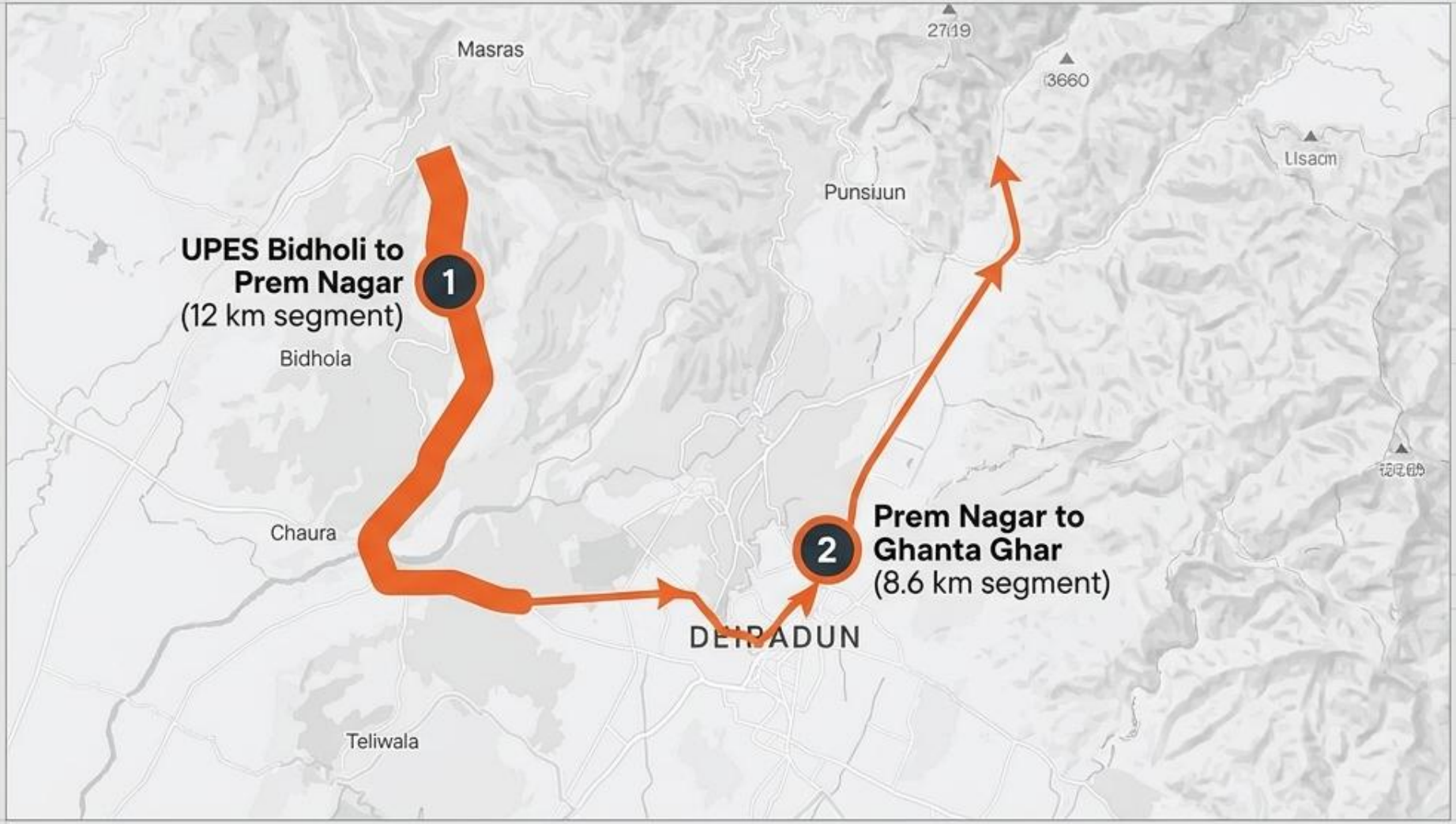
Autonomous transit for Tier 2 & 3 India



Aaryan Haldankar

Mechanical Engineer
Transportation designer

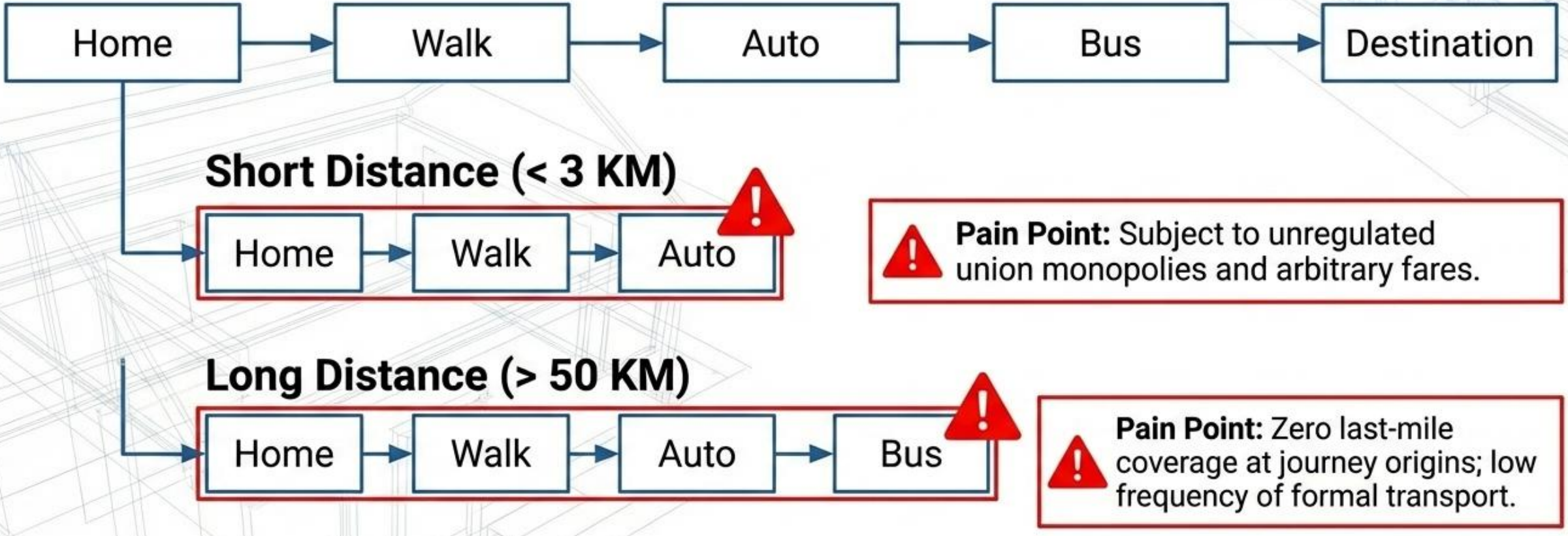
Terrain Constraints and Route Analysis (Dehradun Case Study)



Infrastructural Realities

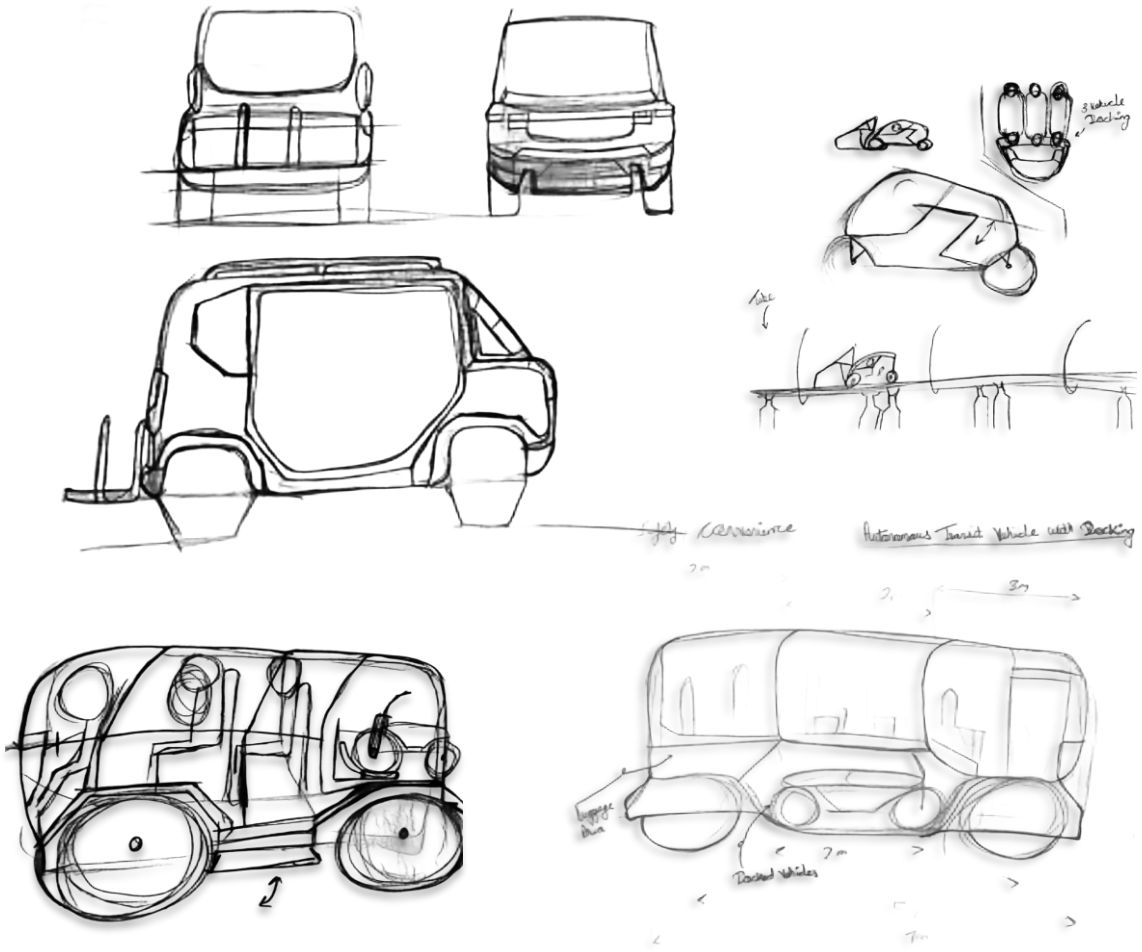
- Smart City road projects delayed; heavy encroachments.
- Narrow road bottlenecks forcing single-file traffic.
- Public transport share dropped from 11.1% (1951) to just 1.1% today.

User Journey Systems Mapping



The lack of an integrated feeder system forces reliance on unsafe, polluting private transport, eroding public trust in the mobility network.

Concept & Form Explorations



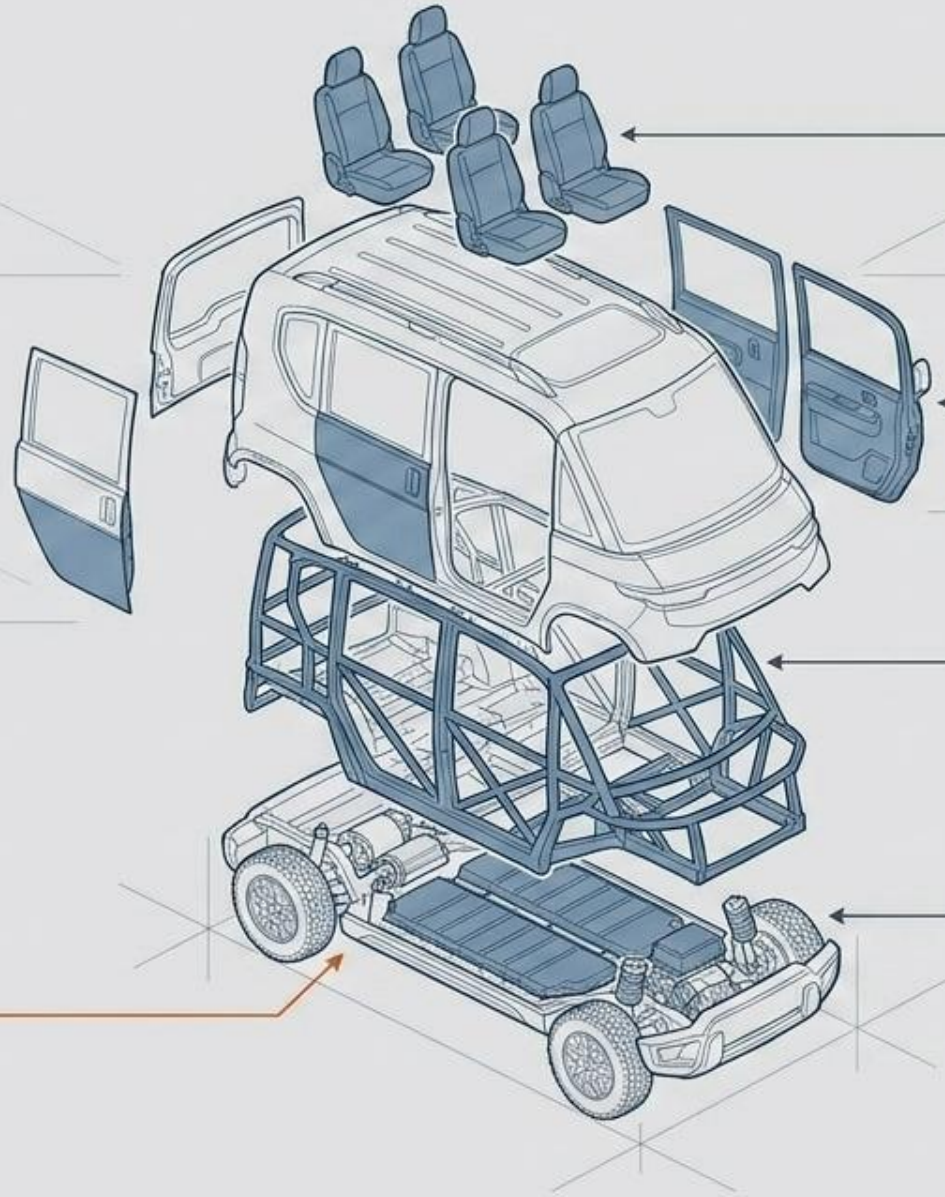








Platform, Materials, and Manufacturing



Layer 4: Modular, wipe-down transit seating

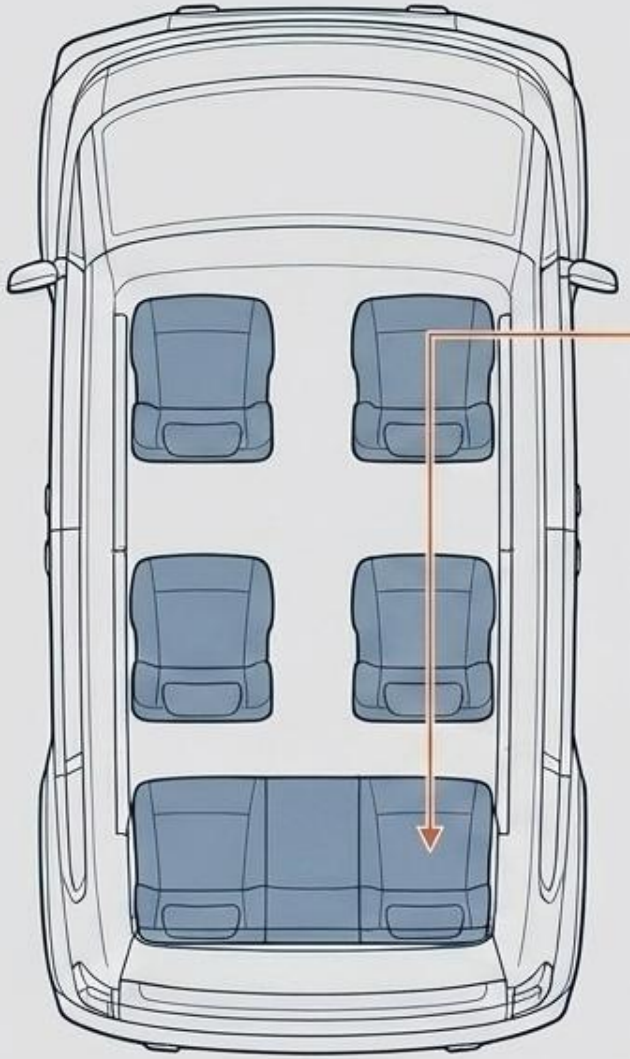
Layer 3: Thermoplastic Body Panels

Layer 2: Steel Tubular Spaceframe

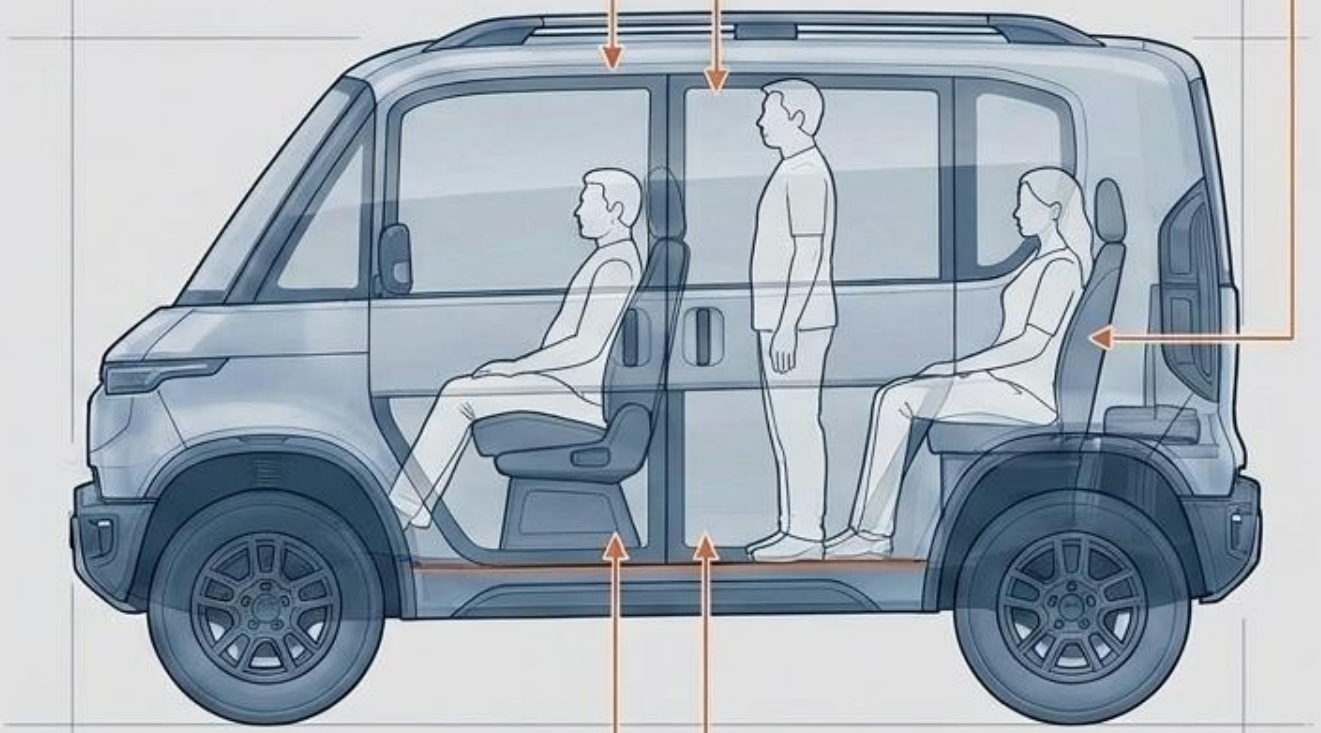
Layer 1: Low-cost EV Skateboard Platform

Fleet-First Maintenance Strategy: Modular thermoplastic panels allow overnight swapping of damaged parts to ensure maximum uptime during the 6 AM–2 AM active window.

Interior Ergonomics and Spatial Packaging



Forward-facing seats exclusively—directly eliminating the motion sickness reported in legacy Tracker facing-seat configurations.



High roofline for comfortable ingress and egress.

Modular rear jump-seats to accommodate extra luggage without penalizing passenger fares.

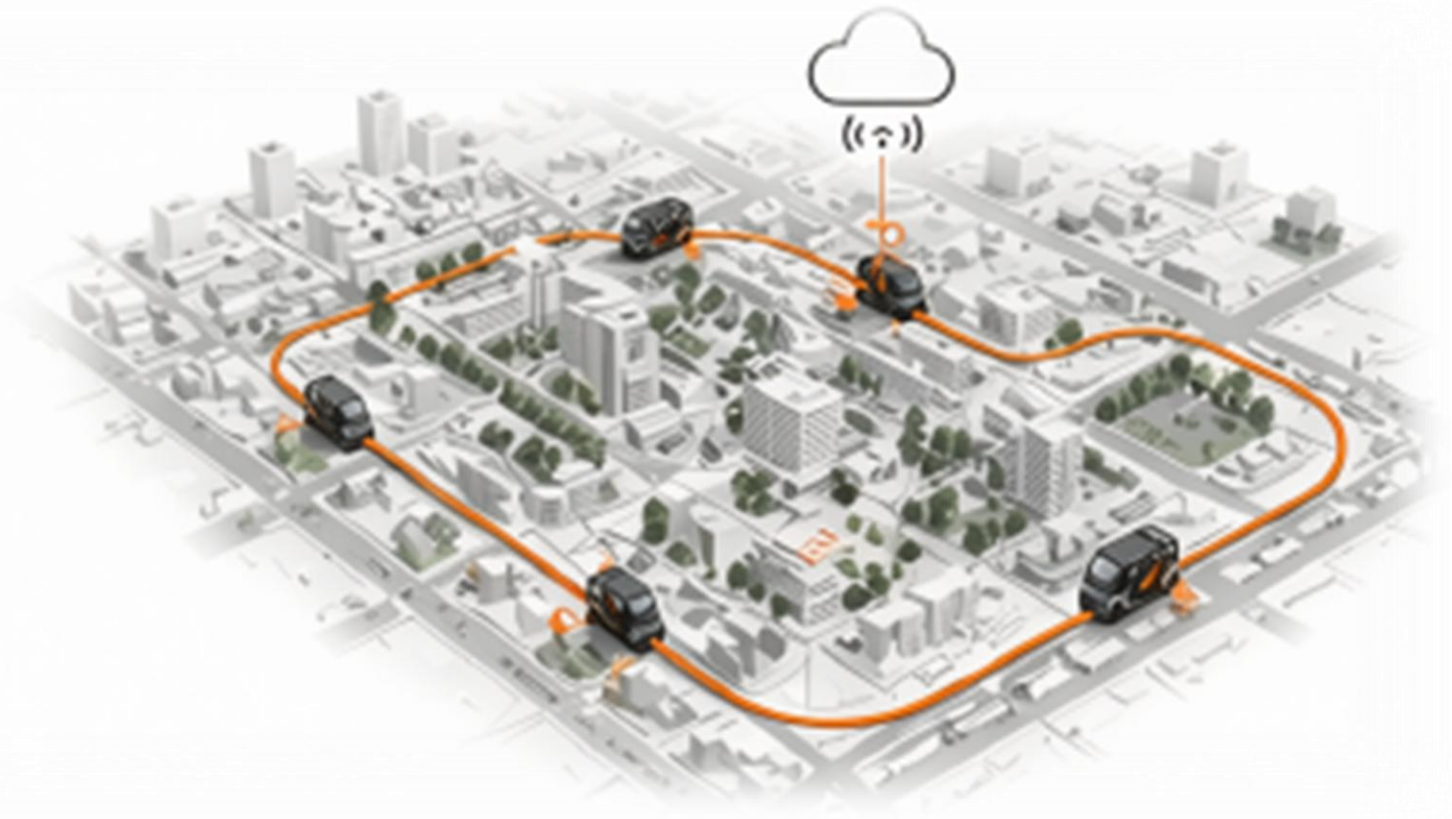
Flat-floor EV platform enabling seamless walk-through.

Modular rear jump-seats to accommodate extra luggage without penalizing passenger fares.

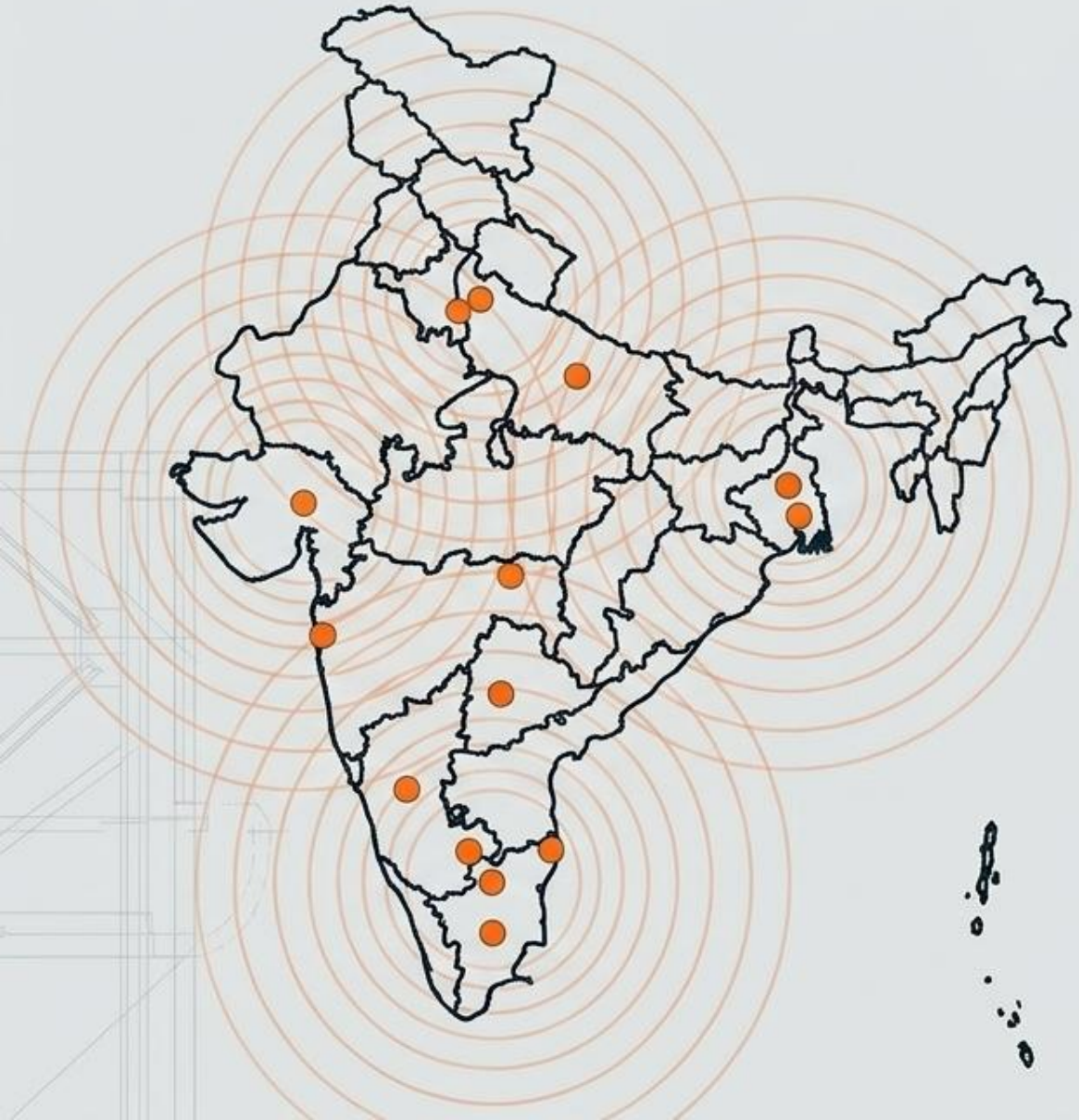
Ecosystem & Deployment

Scalable Deployment

Fixed route autonomous pod working as a connected fleet to ensure punctual reliable and efficient service



National Scalability and Deployment Corridors



- University Campuses
- Sprawling Hospital Networks
- Closed-loop IT Parks
- High-density Tourism Zones
- Gated Residential Communities
- Tier-2 Smart City Core Zones